Project Title: Are taxi rides just a game? Analyzing taxi tipping behavior between drivers and passengers
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Part I: Background and Problem Statement
It is necessary that systems within urban transportation produce high-quality and efficient service despite the pressures of congestion and time, which are especially prominent factors in New York City. Every ride taken will involve high-stakes exchanges between passengers and drivers with partially misaligned incentives. From the passengers' perspective, they would ideally want direct, fast routes that minimize the time and end cost. However, the drivers may face a trade-off between maximizing their meter-based earnings through longer trips or maximizing tips by providing efficient, good customer service. This creates a strategic environment where rational behavior is not directly tied to collective optimal outcomes. The risk factor is that longer trips may come with no tip; on the other hand, quick rides lead to a lower meter. Service quality and group size are recognized as factors influencing the customer's decision-making on tipping amount (Norris et al., 2023). This highlights how observable service characteristics influence tipping behavior, supporting our analysis of the role of trip efficiency on tipping decisions.

This problem needs to be addressed because it has direct welfare and economic implications. Inefficient routing can have several effects, from congestion to an increase in travel time and to costs in an already overburdened urban environment. Tipping is sustained by social norms because it is expected behavior. Cooperation is maintained even when economic incentives are wavering (Kang, 2021). This practice continues to be widespread and contributes to appreciation and cooperation among service providers. Evolutionary game theory and replicator dynamics allow us to model tipping and driver behavior to explain whether tipping is a mechanism that aligns incentives or whether it can break down at any moment due to self-interested strategies. As time evolves, tipping trends may become clear to drivers, and they may begin to favor a certain strategy over the other. This question allows us to observe a real-world setting for how cooperative norms come into existence and persist under conflicting incentives.

Part II: Methods and Expected Results
Using evolutionary game theory and replicator dynamics, we will model the interaction between taxi drivers and customers as a two-population game with misaligned incentives. For the individual customer, it could be considered optimal not to tip when there are no social costs, as this minimizes their costs. For drivers, they choose between meter-maximizing behavior (i.e., longer trips) or tip-maximizing behavior (i.e., more efficient service). These different strategy

combinations determine payoffs, and we are interested in unpacking how these trade-offs and dynamics evolve into an equilibrium, particularly in the rush hour scene of New York City.

Let *x* be the fraction of drivers playing Tip-Max (TM) and *y* the fraction of customers playing Tip (T).

Driver payoffs:

	Tip	No Tip
Tip Maximizing	τ	0
Meter Maximizing	$m + \tau$	m

Customer payoffs:

	Tip Maximizing	Meter Maximizing
Tip	$s_0 + aln(n) - \tau$	-k-τ
No Tip	$-\ \beta ln(n)$	-k

Here τ is the tip, m the extra fare from meter-maxing, k the customer's time cost of a longer trip, s_0 the baseline reciprocity payoff from tipping after efficient service, n the number of passengers, and α and β the per-observer social reward for tipping and social cost of stiffing, respectively. Payoffs are measured relative to a baseline efficient ride. T denotes conditional tipping (tip only when service is efficient), so (T, M) and (N, M) both yield $-k$. Passenger count only enters the customer side: $\alpha \cdot \ln(n)$ rewards tipping in front of observers, $\beta \cdot \ln(n)$ penalizes not tipping, and both vanish in solo rides. We set $\beta = \alpha$ in the baseline and will test $\beta > \alpha$ as a loss-aversion robustness check.

Tipping functions as an informal mechanism that rewards efficient service, while longer trips shift driver earnings toward the meter. We expect larger tip amounts associated with efficient routing, consistent with reciprocity, and higher tip *incidence* in multi-passenger trips, consistent with social signaling. If tipping persists even when efficiency is low, this would suggest a social equilibrium sustains cooperation beyond pure reciprocity, with tipping acting as an evolutionarily stable strategy that incentivizes driver effort. Shifts toward meter-focused driver behavior can be interpreted as mutant strategies invading the cooperative tipping norm, with the stability of (TM, T) depending on whether the tipper share y exceeds the threshold m/τ.

Part III: Significance and Impact

Our results will hopefully allow us to learn more about the games and trade-offs that people deal with and play every single day. Understanding the interactions between taxi drivers and passengers enables a greater understanding of urban efficiency and labor outcomes in not only transportation systems but in the wider world. In New York City, millions of rides occur each day, and understanding the small discrepancies in tipping and driving behavior can be scaled to have profound economic and environmental impacts (Usman, 2024). By viewing tipping through our evolutionary game theory lens, we hope to understand whether the concept of voluntary payments can help align incentives between drivers and passengers or whether no sufficient changes will be made over time.

If we come to find that tipping or the prospect of it encourages better driving behavior, then we could help prove the idea that informal social norms can improve service quality without strict regulation. Using our research, we can help policymakers in places like New York City reinforce this positive tipping culture to help rider and driver relationships. If we find that the concept of tipping and the metered cabs lead to inefficiencies or strategic exploitation from one party, then our research can be used as proof for more formal regulations for compensation. When goals are interrelated, it is imperative that we use a game theory model approach (Norris et al., 2023). On a broader scale, this project looks at an interesting question of how conflicting interests play out in an environment with limited enforcement. The action of a metered cab is repeated and performed by millions of unique individuals who are all placed into an environment where they are partially aligned, but where tension also exists between guaranteed pay and performance-based compensation. Finally, this project aims to find whether tipping can function as an effective coordination mechanism or whether it creates an ecosystem that succumbs to breakdown. By using our models with real-world data, we can enhance ride platforms and policymaking that create a fairer and more sustainable transportation system.

References

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